CALCULUS · UNIT 5: ANTIDERIVATIVES AND DEFINITE INTEGRALS

Solar Energy Audit

Use Riemann sums, antiderivatives, and definite integrals to measure a school's solar energy.

CP1

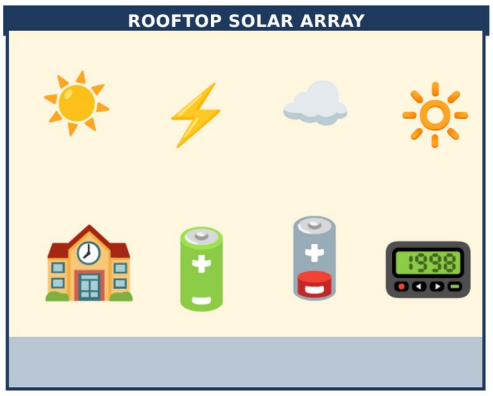
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your school installed solar panels on the roof. A meter records the power output (in kilowatts) every two hours, but the principal wants to know the total energy (in kilowatt-hours) produced each day.

Energy is the accumulation of power over time. You will estimate and calculate it with Riemann sums and integrals.



Units

Power in kW

Energy in kWh

Goal

Cover 600 kWh

per school day

Learning Goals

- I can approximate definite integrals with Riemann and trapezoidal sums.
- I can find antiderivatives and evaluate definite integrals.
- I can use properties of definite integrals.

Project Timeline

When	What you will do
Day 1	Data Sums
Day 2	Antiderivatives & Integrals
Day 3	Upgrade Plan
Final	Energy Audit Report + reflection

Part 1: Data Sums

Day 1 · Use the Power Meter Readings (Δt = 2 h).

Power Meter Readings (kW)

6 a.m.	8 a.m.	10 a.m.	12 p.m.	2 p.m.	4 p.m.	6 p.m.
0	24	58	72	60	30	4

1. Find the left, right, and trapezoidal sums.

2. Use a midpoint sum with 3 intervals of 4 hours (midpoints 8 a.m., 12 p.m., 4 p.m.).

3. Power increases from 6 a.m. to noon. On that interval, is the left sum an over- or under-estimate? Why?

4. Write the energy from 6 a.m. to 6 p.m. as a definite integral using P(t).

Part 2: Antiderivatives & Integrals

Day 2 · $P(t) = 72 - 2(t - 6)^2$, t hours after 6 a.m.

5. Expand P(t) and find an antiderivative.

6. Find the energy from 6 a.m. to 12 p.m. and from 12 p.m. to 6 p.m.

7. Use properties to find $\int_0^{12} P(t)\,dt$ from the two halves.

8. Find $\int_{12}^{0} P(t)\,dt$ and explain the sign.

Part 3: Upgrade Plan

Day 3 · Goal: 600 kWh per day.

9. If each new panel adds 4% to P(t), how many panels are needed so $\int_0^{12} P\,dt \geq 600$?

10. A cloudy day has power 0.6P(t). Find the energy and the shortfall.

11. Write the audit report with a graph, sums, and integrals.

Final Product: Energy Audit Report

Your final product must include:

- Four Riemann-type estimates
- Antiderivative and definite integrals
- Integral properties
- Upgrade plan
- Audit report

Reflection

R1. How do Riemann sums connect to definite integrals?

R2. Why is the midpoint sum often more accurate?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Riemann sums, antiderivatives, and definite integrals are computed correctly and used to judge the energy goal. Goes beyond the minimum with a creative, realistic choice.	Riemann sums, antiderivatives, and definite integrals are computed correctly and used to judge the energy goal.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Audit report includes data, sums, integrals, graphs, and a recommendation. Polished, organized, and visually engaging.	Audit report includes data, sums, integrals, graphs, and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.